IT3021 – Data Warehousing & Business Intelligence

Sri Lanka Institute of Information Technology

Assignment 2

Group : Y3.S1.WE.DS.01.01

Date of submission : 3 April 2026

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Y3.S1.WE.DS.01.01 Batch

Contents

1. Data Source

The data warehouse used for this project, named **DS_Instacart_DW**, was developed based on the Instacart dataset. The dataset was transformed and loaded into a structured data warehouse using a **snowflake schema design** to support efficient analytical processing.

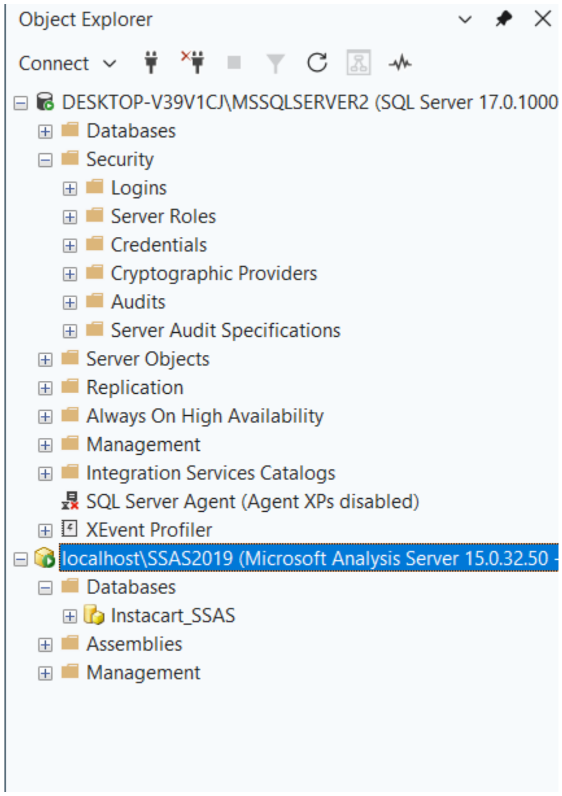


Figure 1: Data Source

The warehouse consists of a central **fact table** and multiple **dimension tables**, enabling multidimensional analysis.

Fact Table

- FactOrder
  - Stores transactional data such as:
    - order_id
    - unit_price
    - discount
    - selling_price
    - txn_process_time_hours

Dimension Tables

The fact table is connected to the following dimensions using surrogate keys:

- **DimCustomer** – contains customer-related information
- **DimProduct** – contains product details
- **DimDate** – contains date attributes for time-based analysis
- **DimRating** – contains product ratings
- **DimAisle** – categorizes products into aisles
- **DimDepartment** – higher-level product categorization

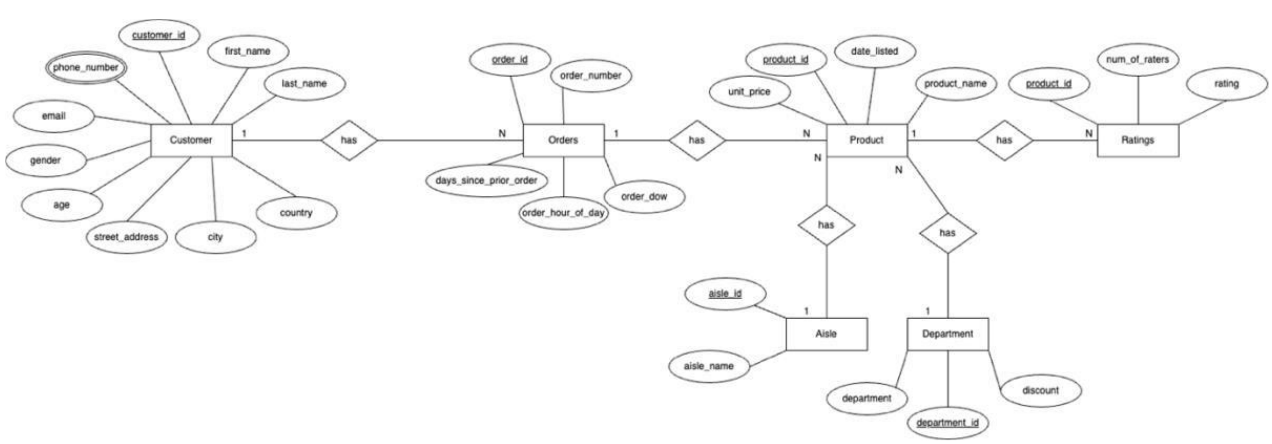
These tables form a **snowflake schema**, where some dimensions

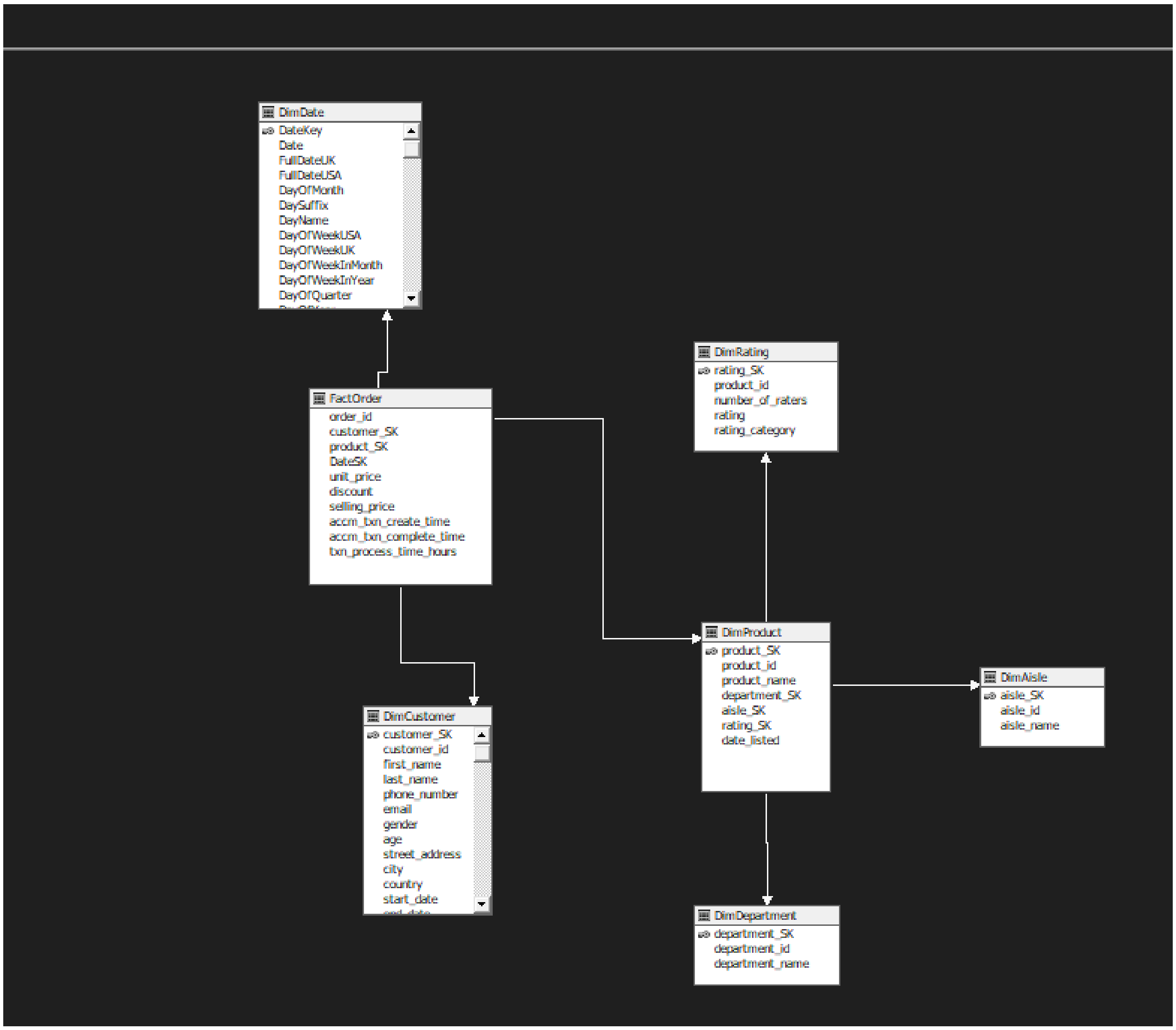
(e.x., Product → Aisle → Department) are further normalized.

This design enables advanced analytical operations such as:

- Time-based analysis (via DimDate)
- Product category analysis (via DimProduct, DimAisle, DimDepartment)
- Customer segmentation (via DimCustomer)

The structured warehouse serves as the data source for building the SSAS cube.




DimDate
DateKey
Date
FullDateUK
FullDateUSA
DayOfMonth
DaySuffix
DayName
DayOfWeekUSA
DayOfWeekUK
DayOfWeekInMonth
DayOfWeekInYear
DayOfQuarter
FactOrder
order_id
customer_SK
product_SK
DateSK
unit_price
discount
selling_price
accm_txn_create_time
accm_txn_complete_time
txn_process_time_hours
DimRating
rating_SK
product_id
number_of_raters
rating
rating_category
DimProduct
product_SK
product_id
product_name
department_SK
aisle_SK
rating_SK
date_listed
DimAisle
aisle_SK
aisle_id
aisle_name
DimCustomer
customer_SK
customer_id
first_name
last_name
phone_number
email
gender
age
street_address
city
country
start_date
DimDepartment
department_SK
department_id
department_name

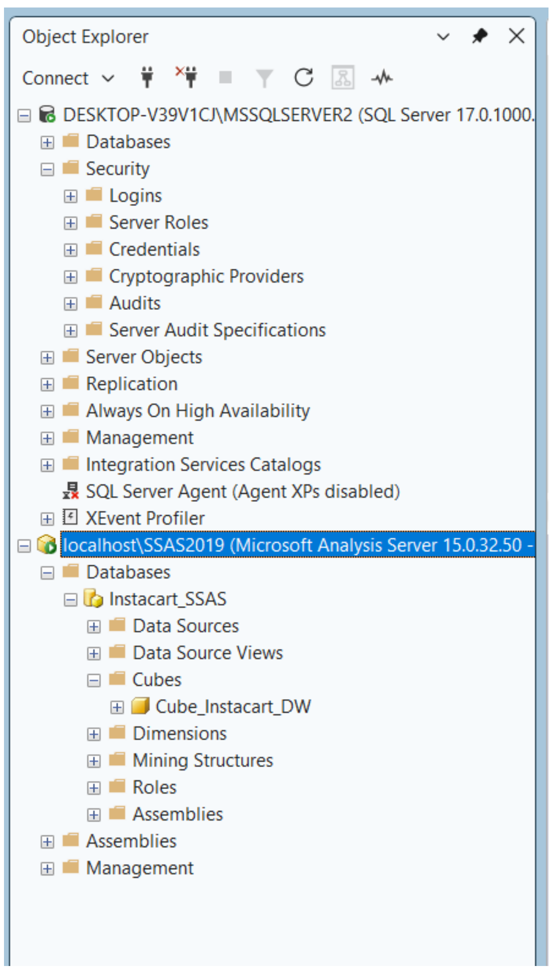
2.SSAS Cube Implementation

A data structure called an OLAP cube also referred to as a multidimensional cube or hypercube enables near real-time data analysis using OLAP databases. It allows users to explore large volumes of data efficiently across multiple dimensions.
In any OLAP cube, two primary components are essential:

- **Dimensions** – Represent descriptive categories (e.g., product, department ,customer, date).

- **Measure Group** – Represents the fact table containing numeric data like sales, discounts, and processing time.

In this project, SQL Server Data Tools (SSDT) was used to create the cube using the steps:
Analysis Services → Multidimensional → Analysis Services Multidimensional and Data Mining Project



2.1 Cube Implementation

2.1.1 Creating the Data Source

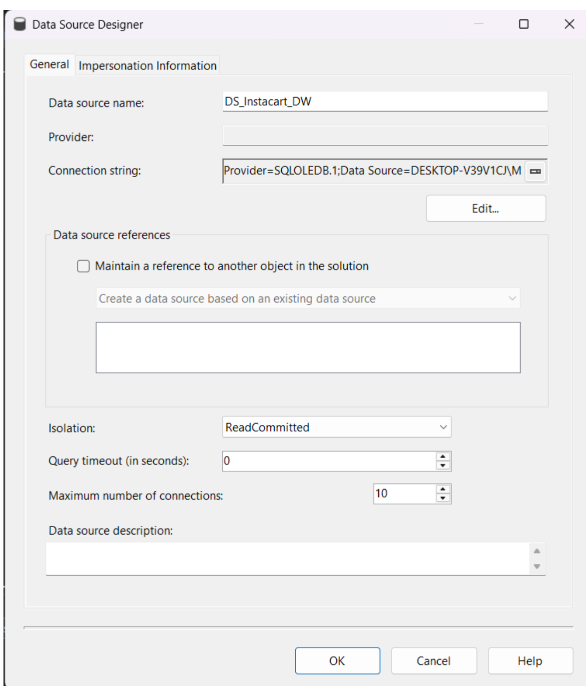
In this scenario, a data warehouse built from the Instacart dataset serves as the data source. The source is defined in the cube project as **DS_Instacart_DW**, as shown in the **Solution Explorer**.

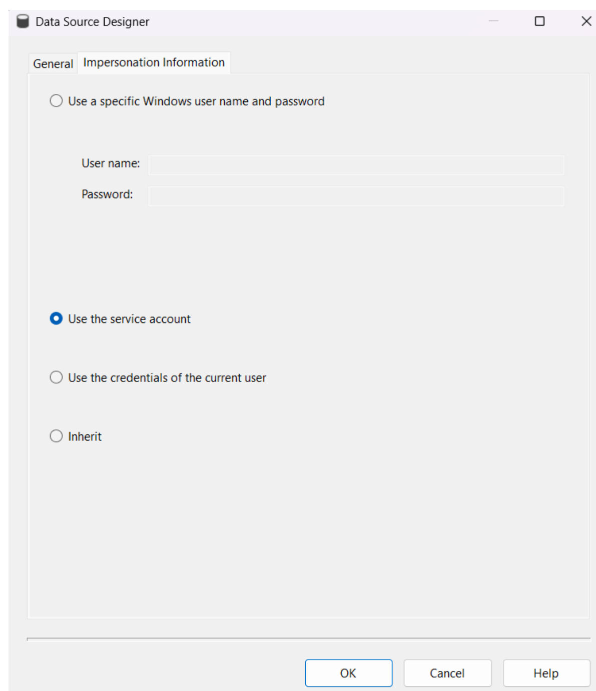
-The cube connects to the SQL Server data warehouse via DS_Instacart_DW using a connection string that includes the local SQL Server instance.

-The data source references the warehouse created from transformed Instacart transactional and dimensional data.

The connection uses:

- Server: localhost\SSAS2019
- Authentication: Windows Authentication





2.1.2 Creating the Data Source View

In SQL Server Analysis Services (SSAS), the **Data Source View (DSV)** acts as a logical layer that defines which tables from the data warehouse are visible to the cube. The analysis service can access only those tables that are included in the DSV. Hence, creating a DSV is a critical step before cube design.

For the **Instacart project**, the DSV was created using the previously defined data source, **DS_Instacart_DW**. After selecting the required fact and dimension tables, the relationships were established based on foreign key references. A proper name was then assigned to the DSV: **DSV_Instacart_DW**.

The DSV includes the following key components:

Tables

* Fact table

FactOrder- Stores transactional data.

*Dimension Tables

DimCustomer – contains customer-related information

DimProduct – contains product details

DimDate – contains date attributes for time-based analysis

DimRating – contains product ratings

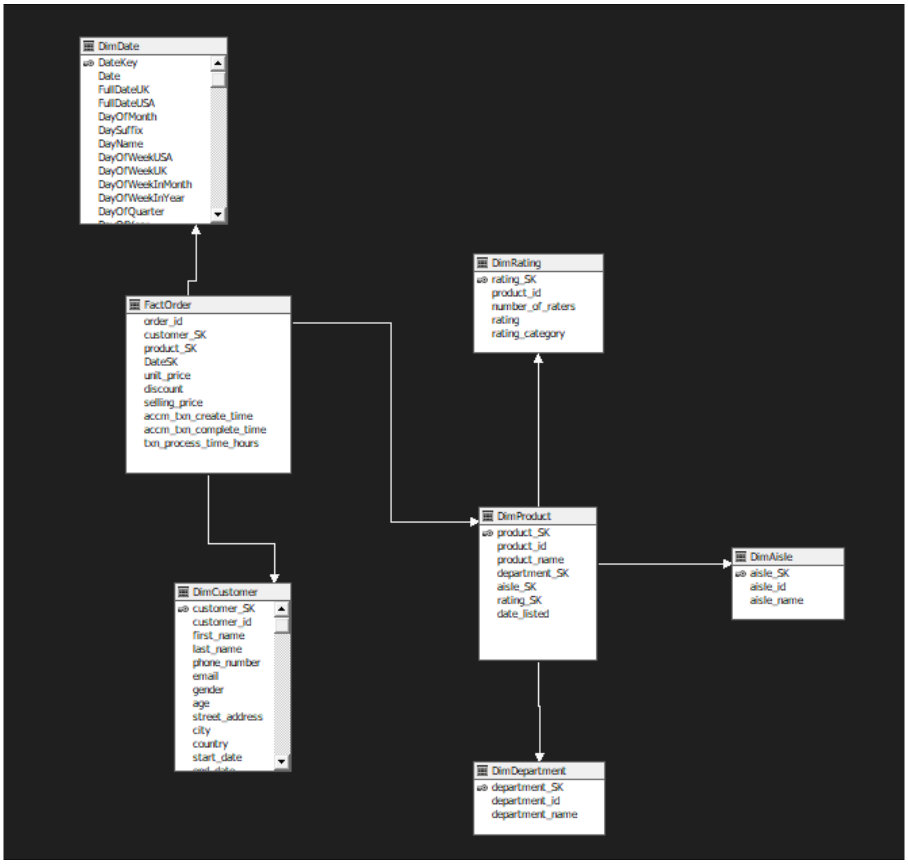
DimAisle – categorizes products into aisles

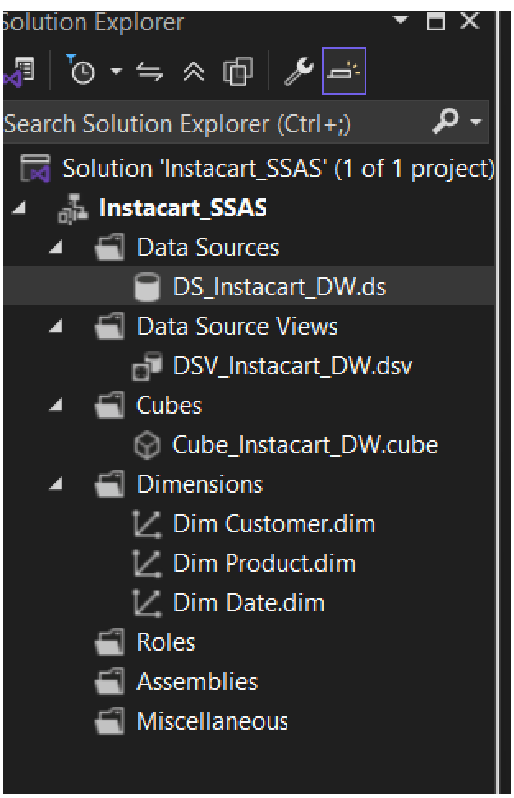
DimDepartment – higher-level product categorization

Relationships

Relationships were automatically detected based on foreign keys

The schema forms a **snowflake structure**.


DimDate
DateKey
Date
FullDateUK
FullDateUSA
DayOfMonth
DaySuffix
DayName
DayOfWeekUSA
DayOfWeekUK
DayOfWeekInMonth
DayOfWeekInYear
DayOfQuarter
FactOrder
order_id
customer_SK
product_SK
DateSK
unit_price
discount
selling_price
accm_txn_create_time
accm_txn_complete_time
txn_process_time_hours
DimRating
rating_SK
product_id
number_of_raters
rating
rating_category
DimProduct
product_SK
product_id
product_name
department_SK
aisle_SK
rating_SK
date_listed
DimAisle
aisle_SK
aisle_id
aisle_name
DimCustomer
customer_SK
customer_id
first_name
last_name
phone_number
email
gender
age
street_address
city
country
start_date
DimDepartment
department_SK
department_id
department_name


Solution Explorer
Search Solution Explorer (Ctrl+;)
Solution 'Instacart_SSAS' (1 of 1 project)
Instacart_SSAS
Data Sources
DS_Instacart_DW.ds
Data Source Views
DSV_Instacart_DW.dsv
Cubes
Cube_Instacart_DW.cube
Dimensions
Dim Customer.dim
Dim Product.dim
Dim Date.dim
Roles
Assemblies
Miscellaneous

2.1.3 Creating the Cube

Using the previously created Data Source View, the cube was implemented through the Cube Wizard in SQL Server Data Tools (SSDT). The wizard guides through selecting the DSV, identifying the fact table as the measure group table, and then choosing the relevant measures and dimensions.

In this scenario, the cube was named Cube_Instacart_DW, and the central fact table selected was FactOrder,

Measure Group:

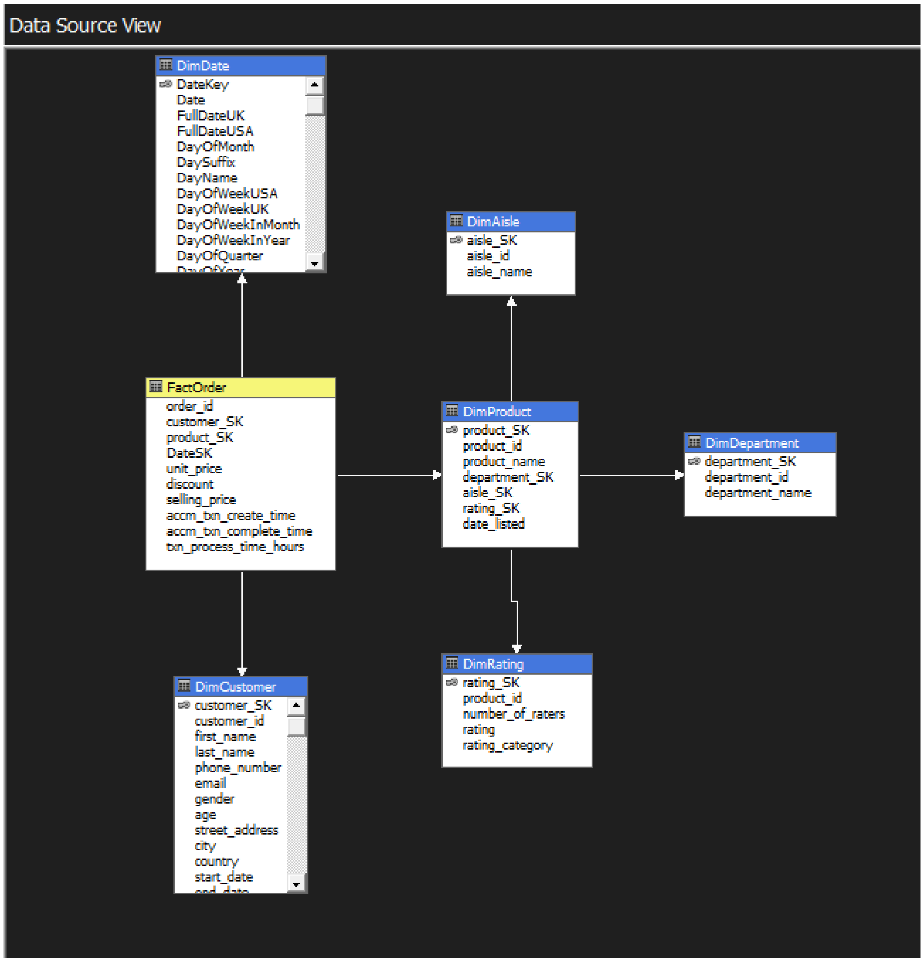
- Fact Order

Measures:

- Order Id
- Unit Price
- Selling Price
- Discount
- Txn Process Time Hours
- **Fact Order Count**

Dimensions:

- DimCustomer
- DimProduct
- DimDate
- DimRating
- DimAisle
- DimDepartment


Data Source View
DimDate
DateKey
Date
FullDateUK
FullDateUSA
DayOfMonth
DaySuffix
DayName
DayOfWeekUSA
DayOfWeekUK
DayOfWeekInMonth
DayOfWeekInYear
DayOfQuarter
DimAisle
aisle_SK
aisle_id
aisle_name
FactOrder
order_id
customer_SK
product_SK
DateSK
unit_price
discount
selling_price
accm_txn_create_time
accm_txn_complete_time
txn_process_time_hours
DimProduct
product_SK
product_id
product_name
department_SK
aisle_SK
rating_SK
date_listed
DimDepartment
department_SK
department_id
department_name
DimRating
rating_SK
product_id
number_of_raters
rating
rating_category
DimCustomer
customer_SK
customer_id
first_name
last_name
phone_number
email
gender
age
street_address
city
country
start_date

2.1.4 Creating Hierarchies and Dimension Structures

Once the cube structure was created, individual **dimensions** appeared in the Dimensions folder of the solution. Attributes from the Data Source View were dragged into the **Attributes pane** of each dimension.
Then, meaningful **hierarchies** were constructed by dragging multiple related attributes into the **Hierarchy pane**, supporting advanced navigation and drill-down within the cube.

DimDate Hierachy

A time-based hierarchy was created with levels:

Year → Quarter → Month → Date

This allows:

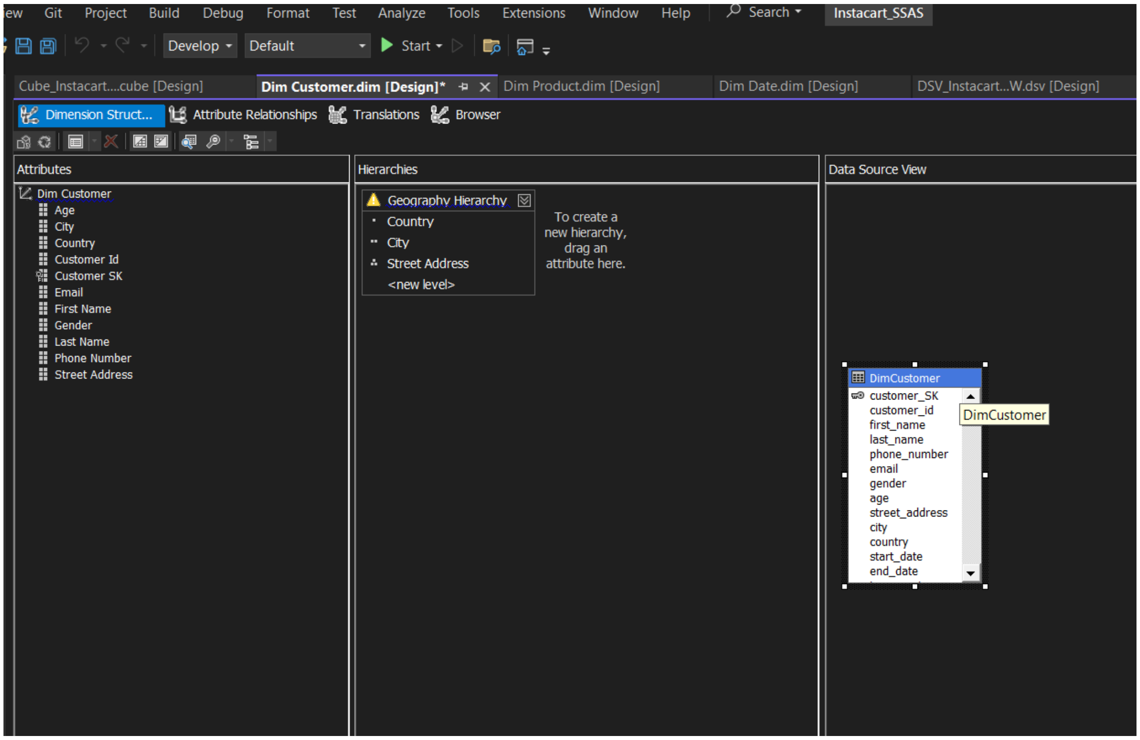
*Aggregation (year level)

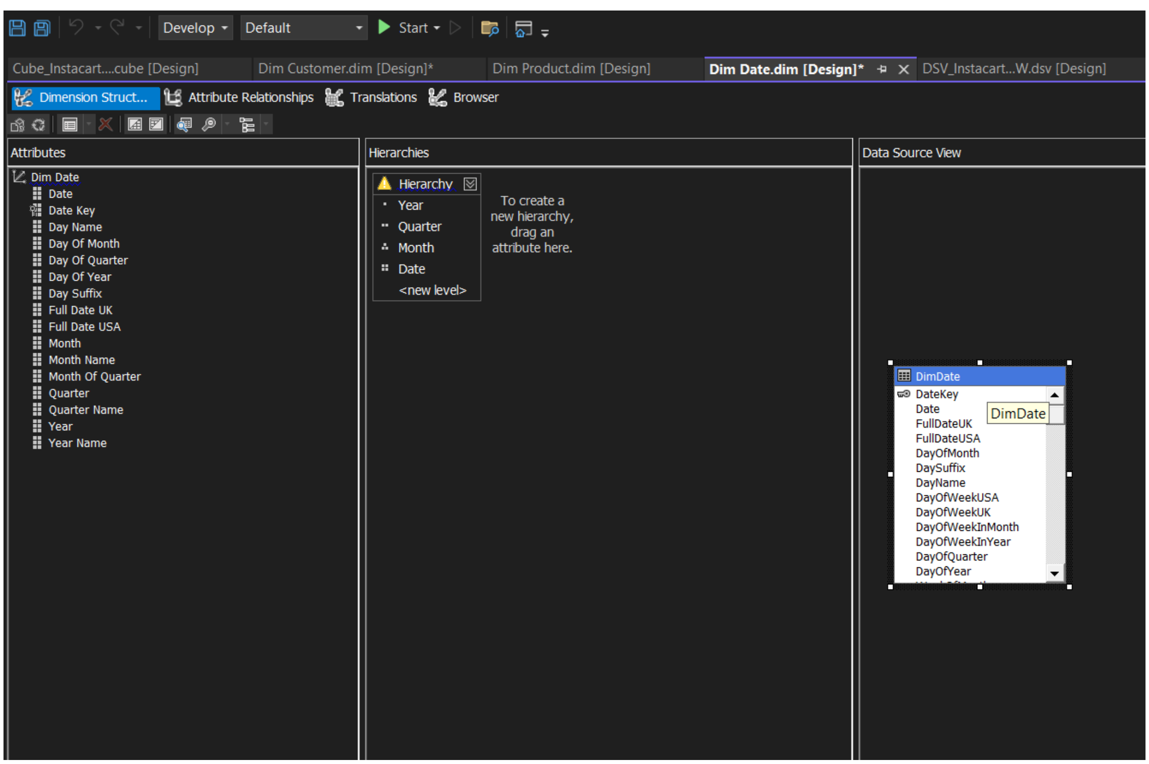
*Drill-down (to months/days)

DimCustomer: A geographic hierarchy was constructed with:

Country → City → Street Address

Enables slicing data by customer location to analyze purchase trends across regions.





2.1.5 Deploying the Cube

Once all cube components—including **dimensions**, **measure groups**, were finalized, the cube was deployed using SQL Server Data Tools (SSDT). The deployment process connects to the target **Analysis Services server** and publishes the cube structure for browsing and analysis.

As shown in the deployment summary, the following steps were executed:

- Processing of the database Instacart_SSAS
- Deployment and processing of the cube Cube_Instacart_DW
- Processing of the Fact Order measure group

3. Demonstration of OLAP Operations

OLAP (Online Analytical Processing) enables users to interactively analyze multidimensional data from multiple perspectives. In this scenario, OLAP operations were performed using **Microsoft Excel connected to the SSAS (SQL Server Analysis Services)** cube built from the Instacart data warehouse.
Once the connection to the SSAS cube was established through Excel's **Data tab**, users were able to carry out various OLAP operations without needing to write MDX queries.

These operations include:

- **Drill Down**: Allowed users to navigate from high-level summaries to more detailed data, such as viewing transaction counts per product, then drilling down to see them by customer.
- **Roll Up**: Enabled aggregation of data, such as analyzing total sales at the monthly level instead of daily.
- **Slice**: Extracted a specific dimension from the cube, such as filtering only a single product or department.
- **Dice**: Selected a sub-cube based on multiple dimensions, for example, analyzing data for a specific product category within a specific time frame.
- **Pivot**: Allowed rotation of axes in the pivot table to change the view of the data, such as swapping rows and columns to view metrics from different perspectives.

These operations provided a dynamic and intuitive way to explore the data and derive meaningful business insights.

3.1 Connecting to the SSAS Cube

To perform OLAP operations in Excel, a connection must be established between the Excel workbook and the SSAS (SQL Server Analysis Services) cube.
This connection allows users to explore cube data interactively using Excel features such as PivotTables, slicers, and filters.

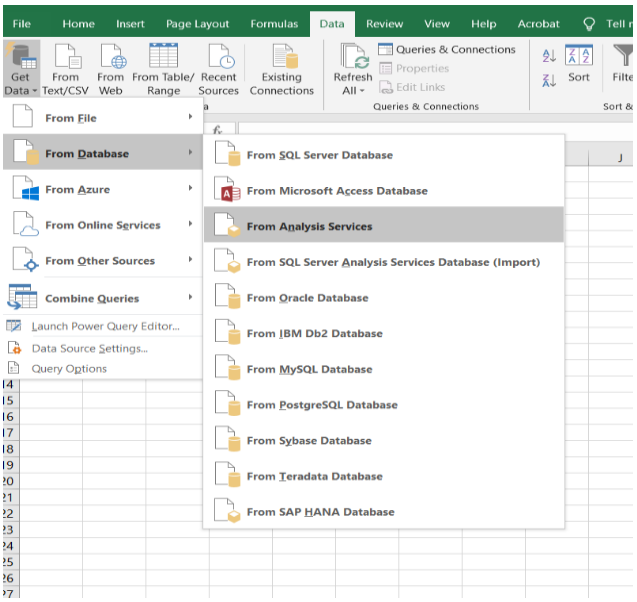
In this scenario, instead of using MDX queries, the **Data tab** feature in Excel was utilized to connect to the cube directly.
The connection process is as follows:

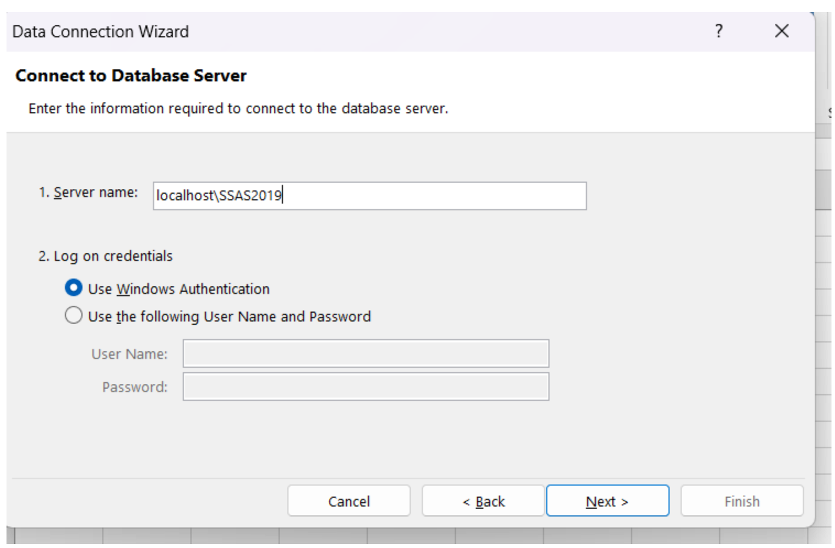
1. Navigate to the **Data** tab in Excel.
2. Select **Get Data > From Database > From Analysis Services**.
3. In the **Data Connection Wizard**
   , provide the
   - **Server** localhost\SSAS2019
4. Use **Windows Authentication** to log in.

5. Chose:

- Database: Instacart_SSAS
- Cube: Cube_Instacart_DW

6. Loaded data as a **PivotTable Report**





3.2. Initial Cube Exploration using Excel PivotTable

After establishing the connection, a PivotTable was created to analyze the cube data. The analysis focused on the **Fact Order Count** measure, which represents the total number of transactions recorded in the FactOrder table.

To perform a time-based analysis, attributes from the **DimDate dimension** were used. Specifically:

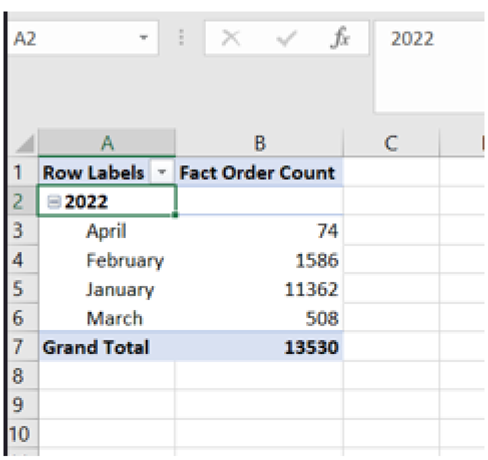
- **Year** was added to the Rows section
- **Month Name** was added below Year to create a hierarchical structure
- **Fact Order Count** was added to the Values section

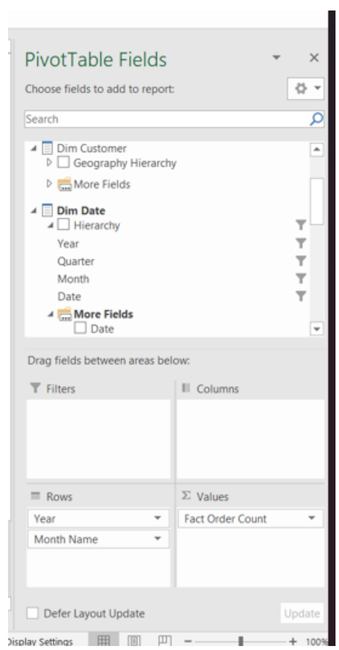
This configuration enabled a hierarchical view of order distribution over time.

The resulting PivotTable shows the breakdown of total orders by month within the year 2022. The results indicate:

- January recorded the highest number of orders
- February and March show a decline in order volume
- April has the lowest number of orders among the displayed months
- The Grand Total represents the total number of transactions for the selected period

This analysis demonstrates how cube data can be aggregated and explored across time dimensions, providing insights into seasonal trends and transaction patterns





3.3 OLAP Operations Demonstration Excel Report

3.3.1. Roll Up

The **Roll Up** operation displayed here aggregates the **Fact Order Count** based on the *aisle* dimension. Instead of analyzing individual product-level transactions, the data is summarized by departments—representing broader product categories.
From the pivot table:

- **Produce** leads with the highest number of orders (996), followed by **dairy eggs** (490), and **beverages** (7984).
- Other categories like **alcohol** (18) and **other** (10) have significantly fewer orders, indicating less user interest or data gaps.
- The **Grand Total** across all aisles is **13530** orders.

1	Row Labels	Fact Order Count
2	⊞ alcohol	18
3	⊞ babies	94
4	⊞ bakery	598
5	⊞ beverages	7984
6	⊞ breakfast	354
7	⊞ canned goods	208
8	⊞ dairy eggs	490
9	⊞ deli	130
10	⊞ dry goods pasta	62
11	⊞ frozen	938
12	⊞ household	72
13	⊞ international	168
14	⊞ meat seafood	176
15	⊞ other	10
16	⊞ pantry	278
17	⊞ personal care	94
18	⊞ pets	174
19	⊞ produce	996
20	⊞ snacks	686
21	**Grand Total**	**13530**
22		
23		
24		

3.3.2 Drill Down

The **Drill Down** operation was performed to explore the data at a more detailed level by navigating through the hierarchical structure of the cube. This operation enables users to move from aggregated summaries to more granular views, revealing deeper insights into the dataset.

In this case, the **Fact Order Count** was drilled down using the **Country level** of the customer geography dimension. Each country represents the number of orders originating from that specific location, allowing for geographical analysis of customer activity.

From the results:

- **Unknown Country dominates with 13,312 orders**, indicating that a large portion of transactions lacks identifiable location data or was not properly recorded.

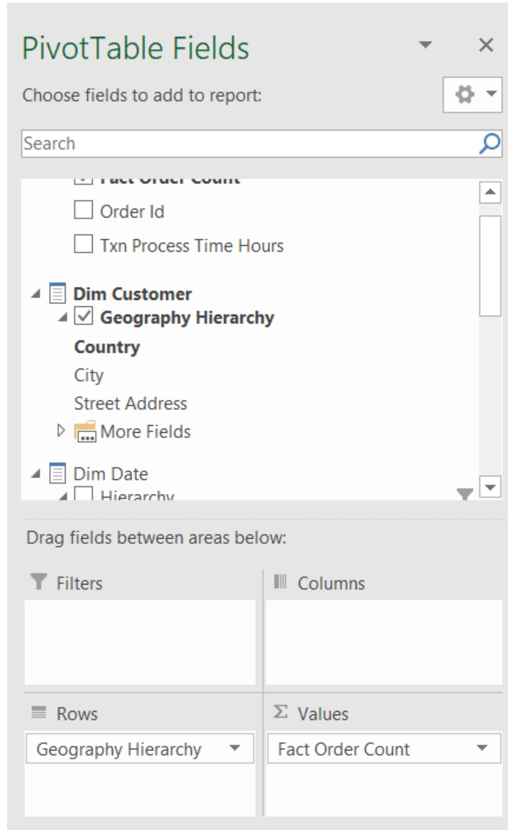
- Among the identifiable countries, **China (54 orders)**, **Indonesia (24 orders)**, and **Malaysia (24 orders)** show relatively higher order volumes.

- Other countries such as **South Africa (16 orders)**, **Russia (8 orders)**, and **Poland (10 orders)** contribute moderate activity.

- Several countries including **Japan, Laos, Latvia, Nigeria, and Venezuela** show minimal order counts, highlighting low engagement or limited data presence.

This drill-down analysis provides valuable insight into the geographical distribution of orders. It helps stakeholders identify regions with strong or weak activity, detect potential data quality issues (such as the high number of unknown locations), and support region-specific decision-making in areas such as marketing strategies, logistics planning, and customer targeting

Row Labels	Fact Order Count
Afghanistan	2
Argentina	4
Bosnia and Herzegovina	2
China	54
Colombia	2
Egypt	4
France	6
Indonesia	24
Iran	2
Japan	2
Laos	2
Latvia	2
Lithuania	8
Malawi	24
Malta	8
Nigeria	2
Peru	2
Poland	10
Portugal	14
Russia	8
Senegal	2
South Africa	16
Sweden	4
Ukraine	2
United States	10
Unknown Country	13312
Venezuela	2
Grand Total	**13530**

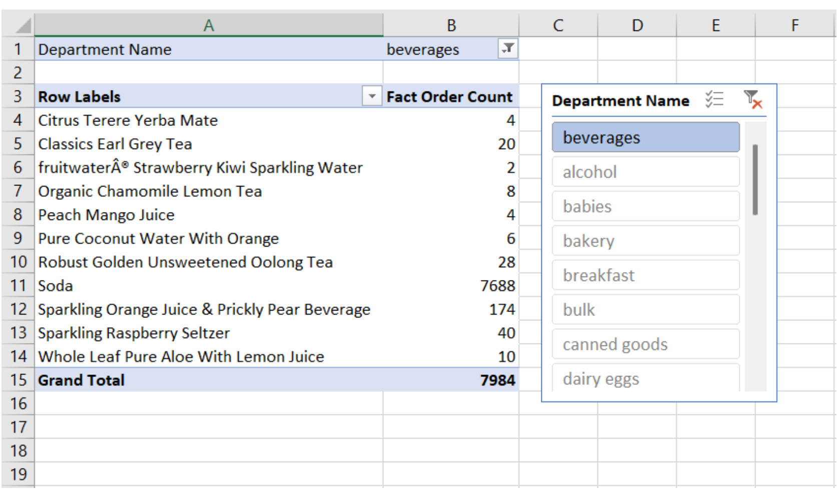


3.3.3.Slice

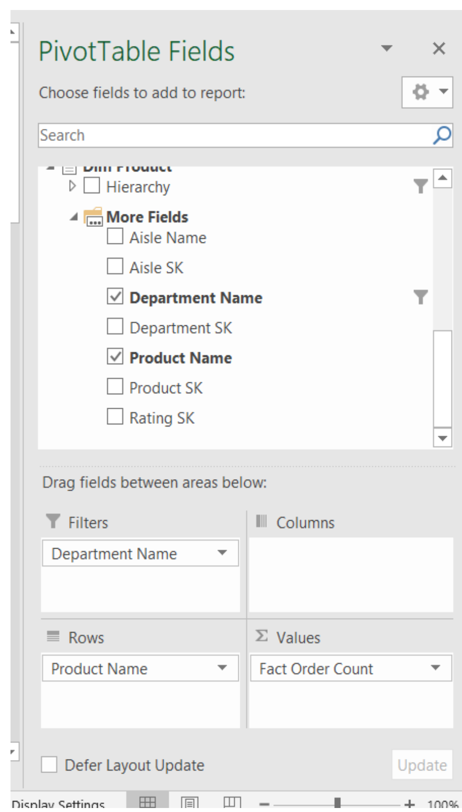
The slice operation was performed using the **Department Name** dimension. The dataset was filtered specifically to the **“beverages”** department, enabling a focused analysis of products within this category. The pivot table presents the **Fact Order Count** for individual beverage products, allowing comparison of product-level performance.

- **Soda** has the highest number of orders (**7688 orders**), clearly dominating the category.
- Other products such as **Sparkling Orange Juice & Prickly Pear Beverage (174 orders)** and **Sparkling Raspberry Seltzer (40 orders)** show comparatively lower demand.

This slicing operation provides detailed product-level insights within a single department. It supports targeted decision-making, such as identifying high-performing products and optimizing category strategies



Department Name	beverages
Row Labels	**Fact Order Count**
Citrus Terere Yerba Mate	4
Classics Earl Grey Tea	20
fruitwaterÂ® Strawberry Kiwi Sparkling Water	2
Organic Chamomile Lemon Tea	8
Peach Mango Juice	4
Pure Coconut Water With Orange	6
Robust Golden Unsweetened Oolong Tea	28
Soda	7688
Sparkling Orange Juice & Prickly Pear Beverage	174
Sparkling Raspberry Seltzer	40
Whole Leaf Pure Aloe With Lemon Juice	10
Grand Total	**7984**

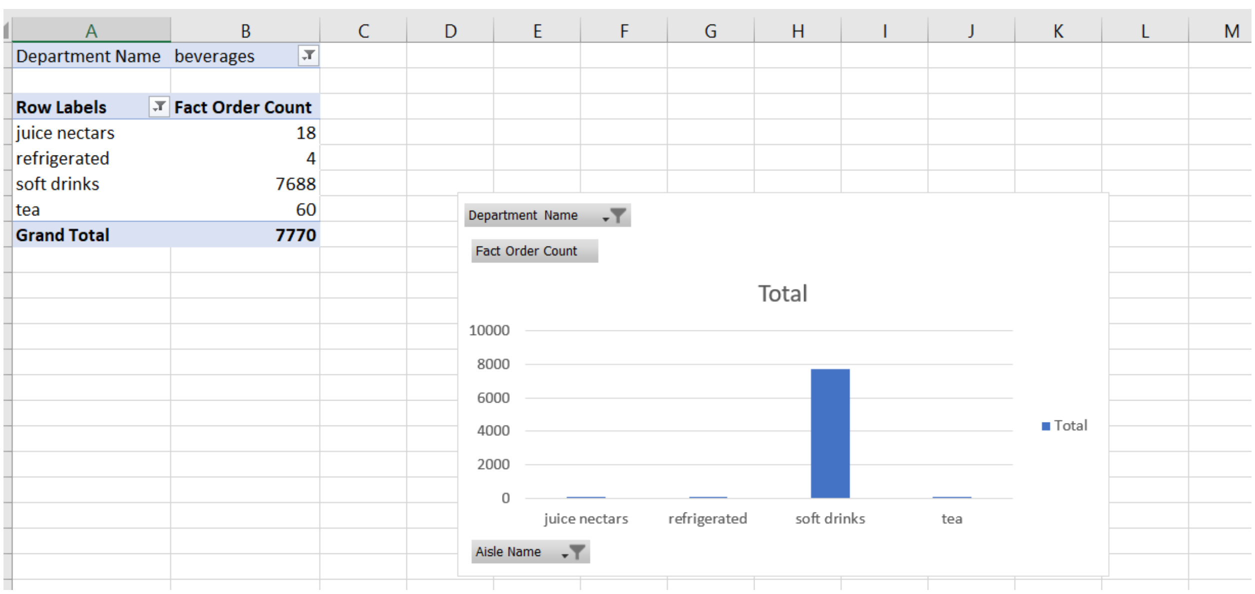


3.3.4Dice

The Dice operation was performed by selecting a subset of data from the cube using multiple dimensional filters. In this case, the dataset was restricted to the **beverages department**, and the **Fact Order Count** was analyzed across selected aisle categories, enabling a more refined multi-dimensional view of the data.

- **Soft drinks** dominate the order volume (**7,688 orders**).
- Other categories such as **tea (60 orders)**, **juice nectars (18 orders)**, and **refrigerated items (4 orders)** show significantly lower demand.

This demonstrates how the Dice operation enables focused analysis of specific segments within the dataset, allowing comparison of performance across multiple related categories



Department Name	beverages
Row Labels	**Fact Order Count**
juice nectars	18
refrigerated	4
soft drinks	7688
tea	60
Grand Total	**7770**

A bar chart was generated to visually represent the distribution of Fact Order Count across selected aisle categories within the beverages department. The visualization clearly shows that *soft drinks* dominate order volume, while other categories such as *tea*, *juice nectars*, and *refrigerated items* contribute comparatively lower counts. This graphical representation enhances the interpretability of the diced data.

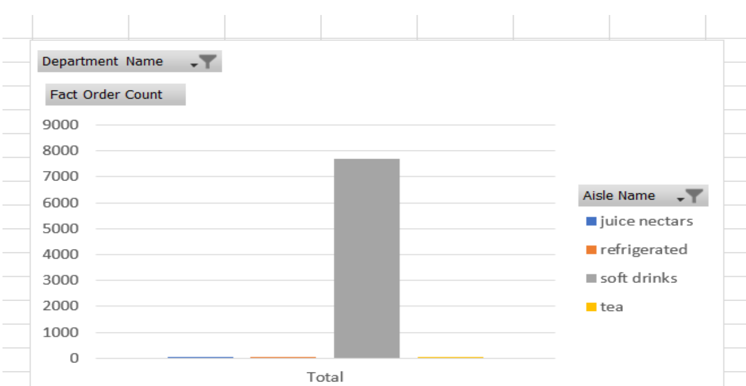
3.3.5Pivot

The **Pivot** operation allows the sub-cube data to be rotated across different dimensional axes to provide alternative views and perspectives of the same measure. This is useful for comparing values by switching row and column fields in the pivot table.

This transformation enables easier side-by-side comparison of different aisle categories within the beverages department. For example, *soft drinks* clearly dominate with the highest order count, while categories such as *tea*, *juice nectars*, and *refrigerated items* show comparatively lower values.

This pivot technique mirrors the concept shown in the initial image, where claimed amounts are analyzed from various angles like year, insurance type, and state—providing multi-dimensional insight through-visual-re-orientation.

	A	B	C	D	E	F
2	Department Name	beverages				
3						
4		Column Labels				
5		juice nectars	refrigerated	soft drinks	tea	Grand Total
6	Fact Order Count	18	4	7688	60	7770



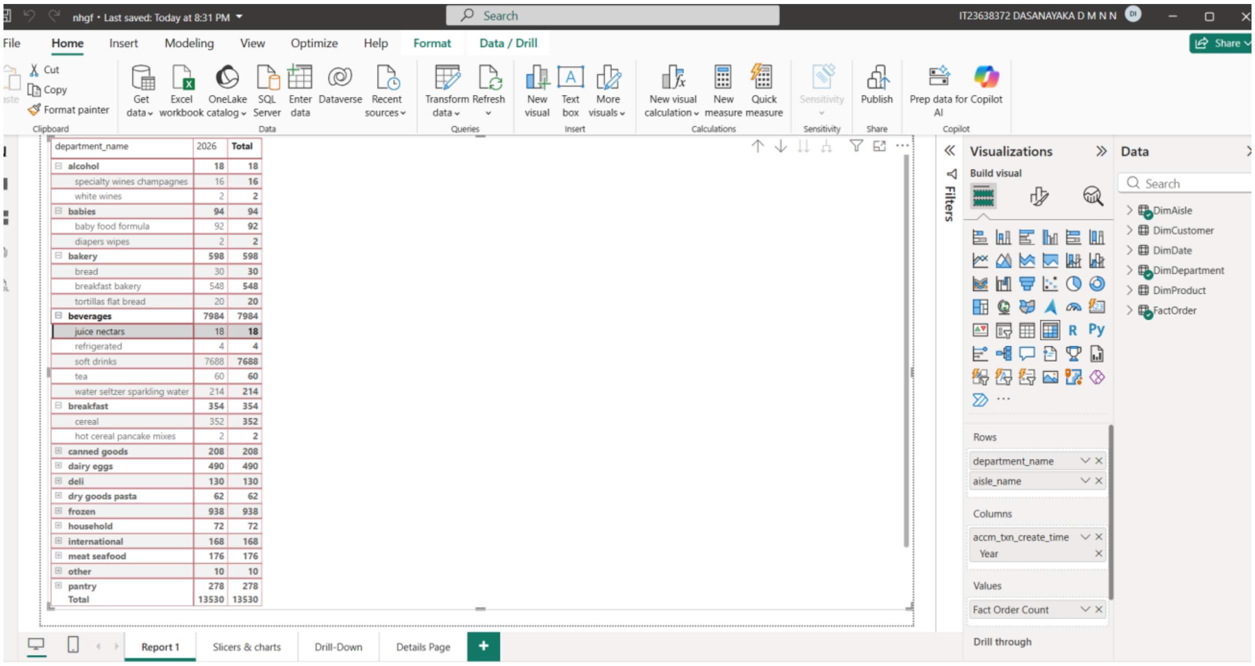
4. PowerBI Reports

4.1 Report 1: Matrix Visual Report – Tabular Display with Groupings

Objective:
To present grouped tabular data using a matrix visual that enables hierarchical analysis across categories and time.

Implemetation

- **Visual Used:** Matrix visual.
- **Data Fields:**
  - Rows: *Department Name*, *Aisle Name*
  - Columns: *Year*
  - Values: *Fact Order Count*
- **Grouping Structure:**
  - Hierarchical row grouping using Department and Aisle.
  - Time-based column grouping using Year.
- **Functionality:**
  - Enables detailed row-level analysis of product categories.
  - Provides a structured comparison of order distribution across departments and aisles over time.
- **Insight:**
  - Offers a compact and organized summary, making it easier to evaluate category-level performance



nhgf • Last saved: Today at 8:31 PM
Search
IT23638372 DASANAYAKA D M N N
File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Share
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Publish
Share
Prep data for AI
Copilot
department_name 2026 Total
alcohol 18 18
specialty wines champagnes 16 16
white wines 2 2
babies 94 94
baby food formula 92 92
diapers wipes 2 2
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bread 30 30
breakfast bakery 548 548
tortillas flat bread 20 20
beverages 7984 7984
juice nectars 18 18
refrigerated 4 4
soft drinks 7688 7688
tea 60 60
water seltzer sparkling water 214 214
breakfast 354 354
cereal 352 352
hot cereal pancake mixes 2 2
canned goods 208 208
dairy eggs 490 490
deli 130 130
dry goods pasta 62 62
frozen 938 938
household 72 72
international 168 168
meat seafood 176 176
other 10 10
pantry 278 278
Total 13530 13530
Filters
Visualizations
Build visual
Rows
department_name
aisle_name
Columns
accm_txn_create_time
Year
Values
Fact Order Count
Drill through
Data
Search
DimAisle
DimCustomer
DimDate
DimDepartment
DimProduct
FactOrder
Report 1
Slicers & charts
Drill-Down
Details Page

4.2 Report 2: Slicers with Cascading Filters

Objective:

To demonstrate interactive data exploration using slicers and multiple visualizations with cascading filtering.

Implementation:

- **Primary Control:**
  - Slicer created using *Department Name* to filter product categories dynamically (e.g., beverages, bakery, dairy).
- **Visuals Used:**
  - **Table Visual:** Displays *Department Name* and *Product Name* for detailed product-level analysis.
  - **Bar Chart:** Shows *Fact Order Count by Product Name* for comparing product performance.
  - **Pie Chart:** Illustrates distribution of orders by *Aisle Name*.
  - **Horizontal Bar Chart:** Displays total orders by *Department Name* to highlight popular categories.
- **Slicer Interaction:**
  - The slicer filters all visuals simultaneously.
  - Selection of a department dynamically updates products, aisles, and metrics across the report.
- **Functionality:**
  - Demonstrates **cascading filtering**, where one selection influences multiple visuals.
  - Supports intuitive and interactive data-driven analysis.
- **Insight:**
  - Enhances user experience by enabling dynamic exploration and better decision-making based on filtered views.

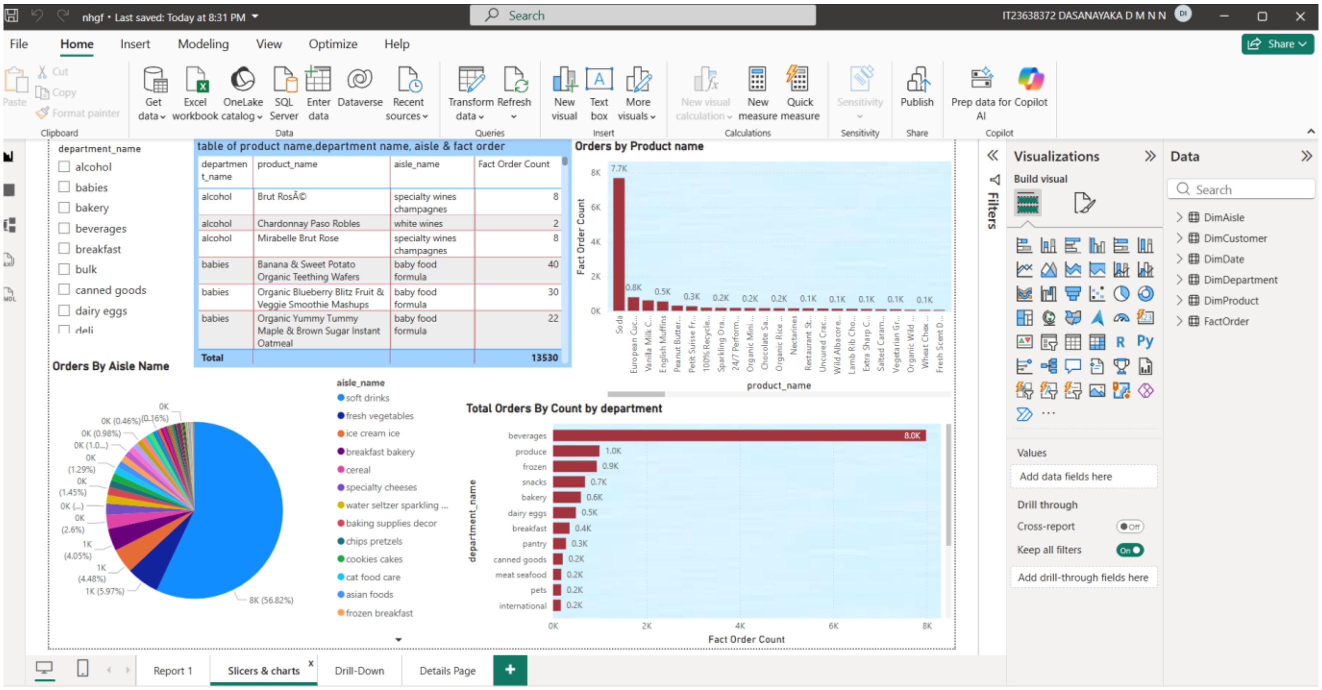
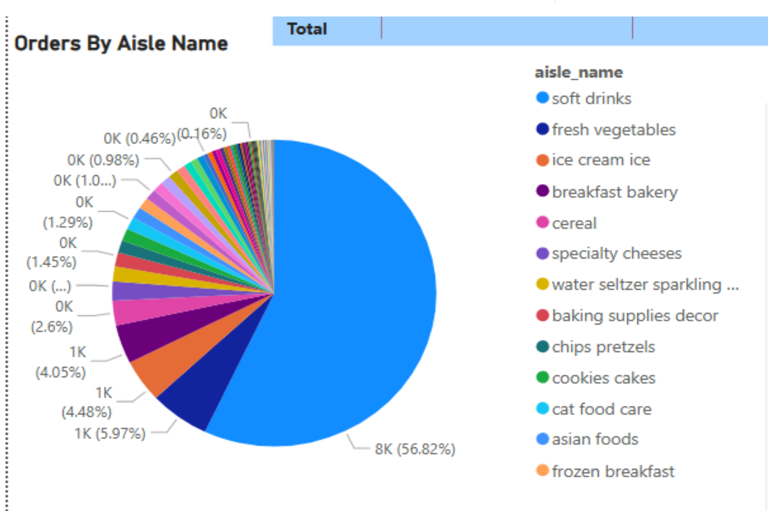
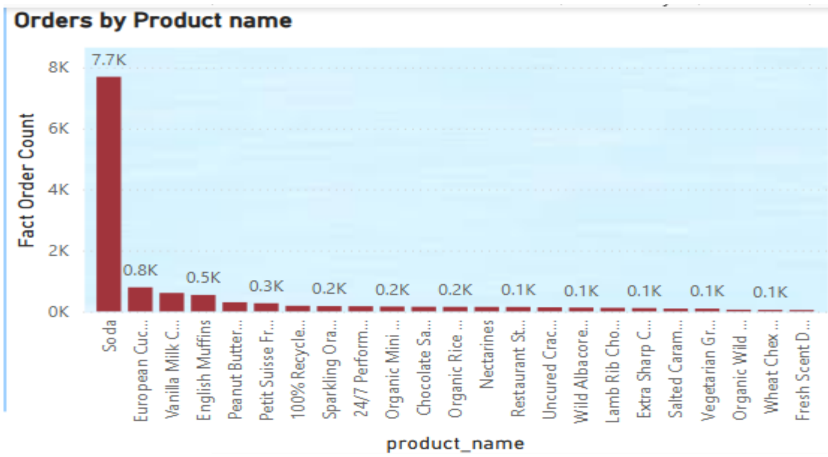
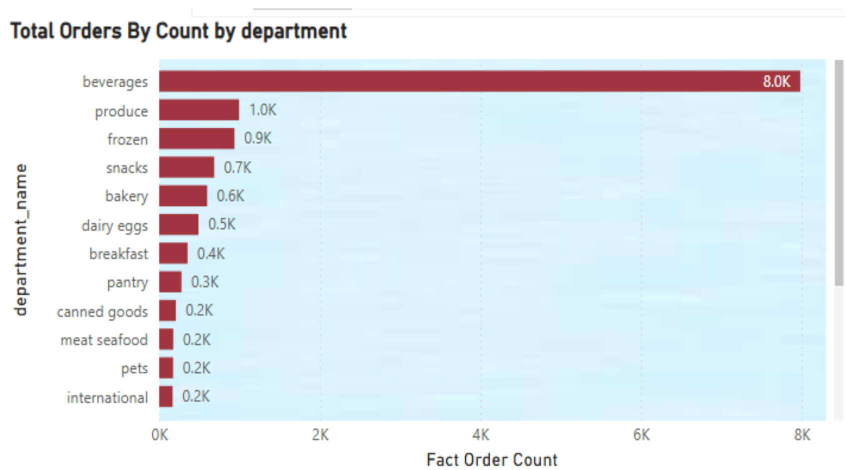


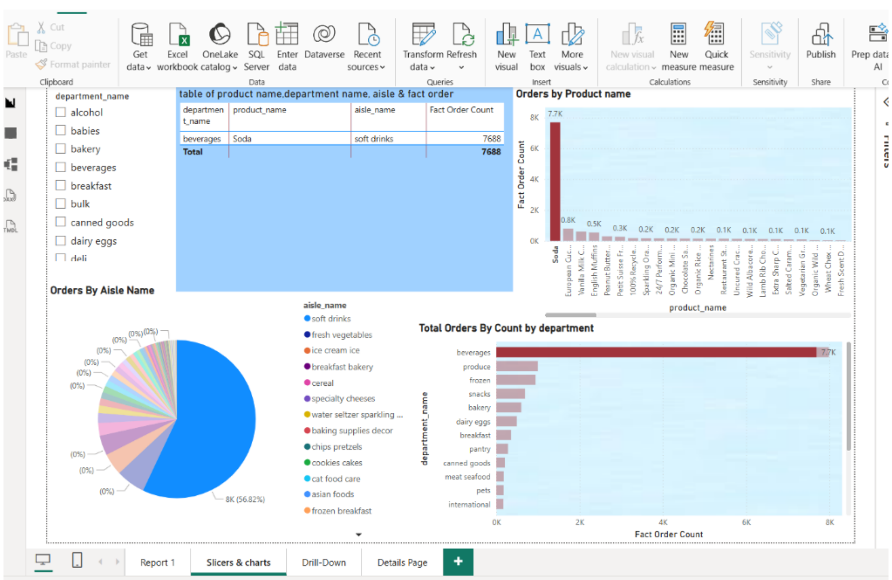
table of product name,department name, aisle & fact order

department_name	product_name	aisle_name	Fact Order Count
alcohol	Brut RosÃ©	specialty wines champagnes	8
alcohol	Chardonnay Paso Robles	white wines	2
alcohol	Mirabelle Brut Rose	specialty wines champagnes	8
babies	Banana & Sweet Potato Organic Teething Wafers	baby food formula	40
babies	Organic Blueberry Blitz Fruit & Veggie Smoothie Mashups	baby food formula	30
babies	Organic Yummy Tummy Maple & Brown Sugar Instant Oatmeal	baby food formula	22
Total			**13530**




Orders by Product name
Fact Order Count
8K
6K
4K
2K
0K
7.7K
0.8K
0.5K
0.3K
0.2K
0.2K
0.2K
0.1K
0.1K
0.1K
0.1K
0.1K
Soda
European Cuc...
Vanilla Milk C...
English Muffins
Peanut Butter...
Petit Suisse Fr...
100% Recycle...
Sparkling Ora...
24/7 Perform...
Organic Mini ...
Chocolate Sa...
Organic Rice ...
Nectarines
Restaurant St...
Uncured Crac...
Wild Albacore...
Lamb Rib Cho...
Extra Sharp C...
Salted Caram...
Vegetarian Gr...
Organic Wild ...
Wheat Chex ...
Fresh Scent D...
product_name


Total Orders By Count by department
department_name
beverages
produce
frozen
snacks
bakery
dairy eggs
breakfast
pantry
canned goods
meat seafood
pets
international
8.0K
1.0K
0.9K
0.7K
0.6K
0.5K
0.4K
0.3K
0.2K
0.2K
0.2K
0.2K
0K
2K
4K
6K
8K
Fact Order Count


department_name
alcohol
babies
bakery
beverages
breakfast
bulk
canned goods
dairy eggs
table of product name,department name, aisle & fact order
beverages
Soda
soft drinks
7688
Total
7688
Orders by Product name
Orders By Aisle Name
aisle_name
soft drinks
fresh vegetables
ice cream ice
breakfast bakery
cereal
specialty cheeses
water seltzer sparkling ...
baking supplies decor
chips pretzels
cookies cakes
cat food care
asian foods
frozen breakfast
Total Orders By Count by department
Fact Order Count
Report 1
Slicers & charts
Drill-Down
Details Page

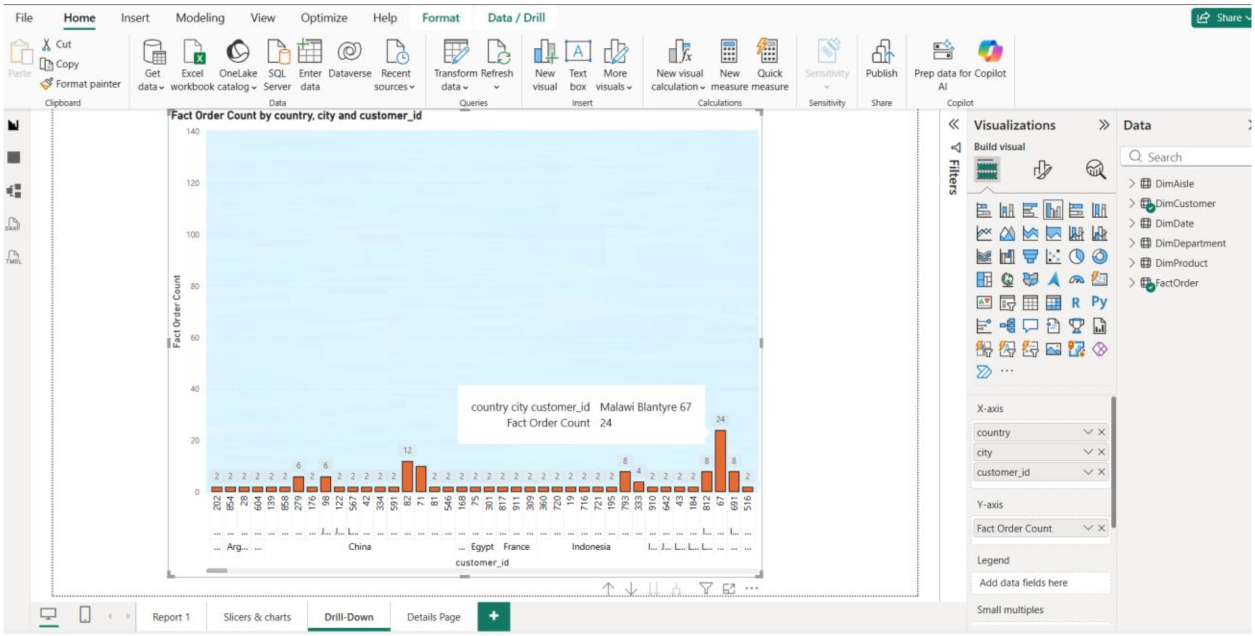
4.3 Report 3: Drill-Down Report

Objective:

To enable hierarchical exploration of data using geographical dimensions for deeper analytical insights.

Implementation:

- **Hierarchy Defined:**
  - *Country → City → Customer ID*
- **Visual Used:**
  - Column chart.
- **Data Fields:**
  - Axis (Hierarchy): *Country, City, Customer ID*
  - Values: *Fact Order Count*
- **Drill-Down Functionality:**
  - Enabled to allow navigation through hierarchy levels.
  - Users can click on a **Country** to view corresponding **Cities**, and further drill down to **Customer ID** level.
- **Reason for Approach:**
  - A geographical hierarchy was used instead of a time-based hierarchy due to limited variation in the date dimension.
- **Functionality:**
  - Supports progressive exploration from aggregated data to detailed customer-level insights.
  - Helps analyze how orders are distributed across regions and customers.
- **Insight:**
  - Provides valuable understanding of **regional performance**, **customer distribution**, and **demand patterns**, supporting location-based decision-making.


File
Home
Insert
Modeling
View
Optimize
Help
Format
Data / Drill
Fact Order Count by country, city and customer_id
Fact Order Count
customer_id
country city customer_id Malawi Blantyre 67
Fact Order Count 24
China
Egypt
France
Indonesia
Visualizations
Build visual
Data
DimAisle
DimCustomer
DimDate
DimDepartment
DimProduct
FactOrder
X-axis
country
city
customer_id
Y-axis
Fact Order Count
Legend
Add data fields here
Small multiples
Filters
Report 1
Slicers & charts
Drill-Down
Details Page

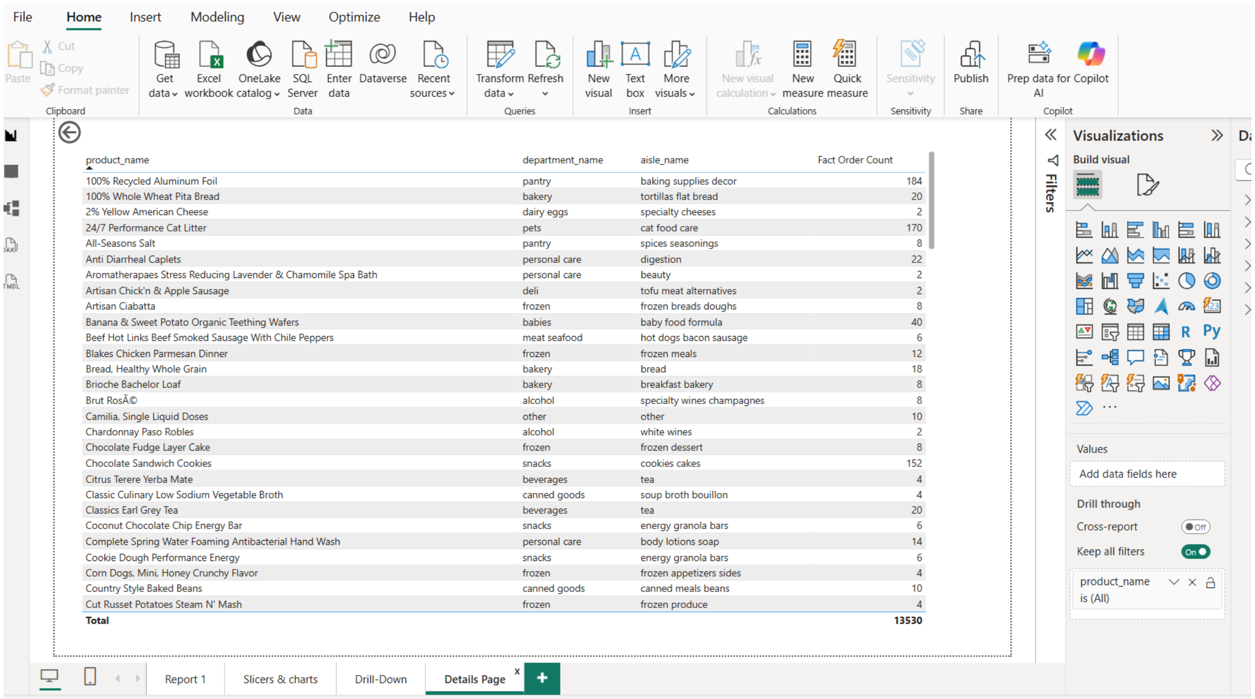
4.4 Report 4: Drill-Through Report

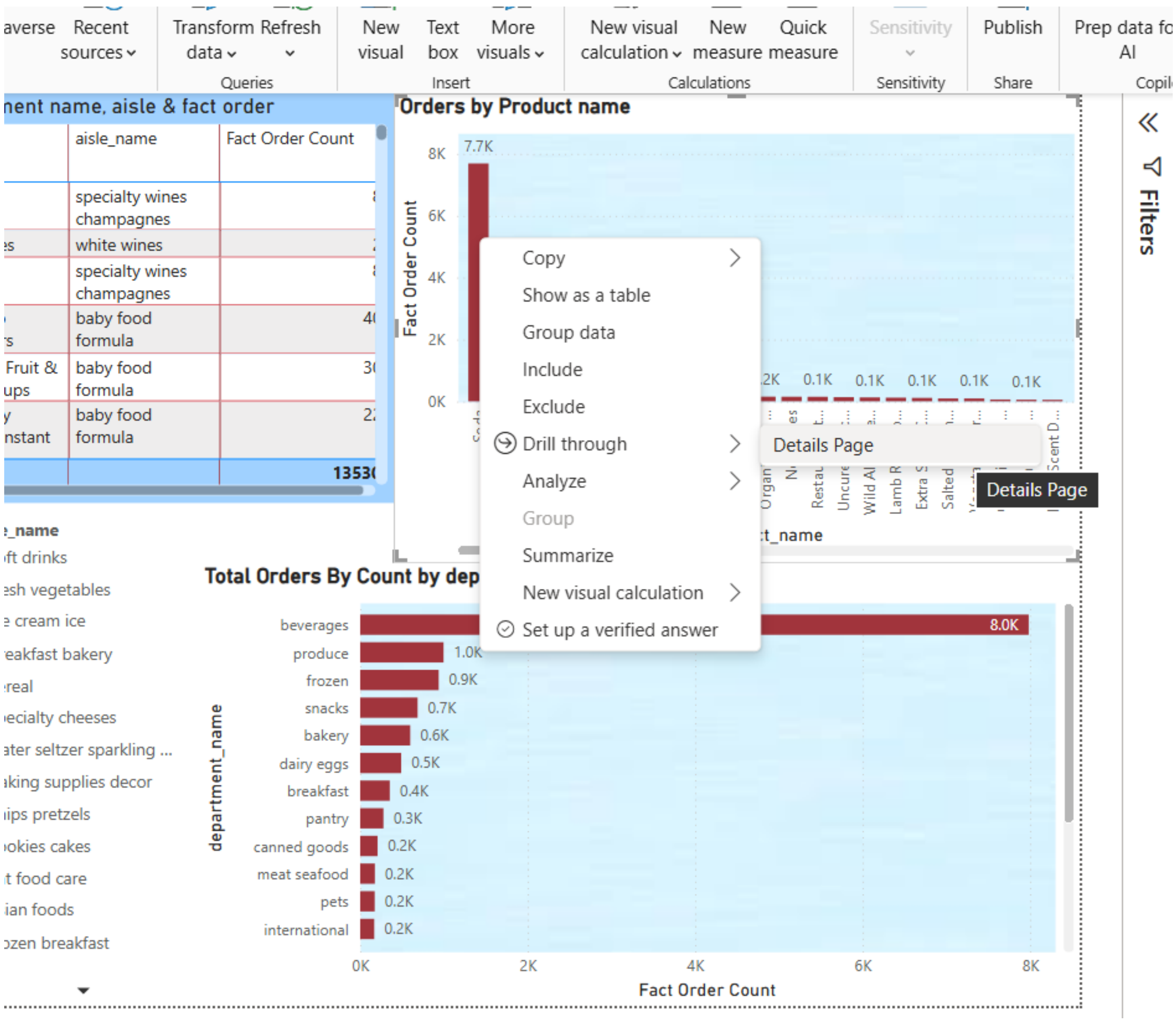
Objective:

To enable detailed analysis of specific data points by navigating from summary visuals to a dedicated detail view.

Implementation:

- **Drill-Through Setup:**
    - A separate page named **“Details Page”** was created.
    - The field *Product Name* was added to the **drill-through filter** section.
    - Users can right-click on a product in another visual to navigate to this page.
- **Visual Used:**
    - Table visual.
- **Data Fields Displayed:**
    - *Product Name*
    - *Department Name*
    - *Aisle Name*
    - *Fact Order Count*
- **Functionality:**
    - Enables users to explore detailed information related to a selected product.
    - Example: Selecting **“Soda”** allows navigation to a page showing its detailed distribution across categories.
- **Insight:**
    - Enhances analytical capability by allowing seamless transition from **summary-level insights** to **detailed data exploration**.


File
Home
Insert
Modeling
View
Optimize
Help
Paste
Cut
Copy
Format painter
Clipboard
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Data
Transform data
Refresh
Queries
New visual
Text box
More visuals
Insert
New visual calculation
New measure
Quick measure
Calculations
Sensitivity
Sensitivity
Publish
Share
Prep data for AI
Copilot
Copilot
product_name
department_name
aisle_name
Fact Order Count
100% Recycled Aluminum Foil pantry baking supplies decor 184
100% Whole Wheat Pita Bread bakery tortillas flat bread 20
2% Yellow American Cheese dairy eggs specialty cheeses 2
24/7 Performance Cat Litter pets cat food care 170
All-Seasons Salt pantry spices seasonings 8
Anti Diarrheal Caplets personal care digestion 22
Aromatherapaes Stress Reducing Lavender & Chamomile Spa Bath personal care beauty 2
Artisan Chick'n & Apple Sausage deli tofu meat alternatives 2
Artisan Ciabatta frozen frozen breads doughs 8
Banana & Sweet Potato Organic Teething Wafers babies baby food formula 40
Beef Hot Links Beef Smoked Sausage With Chile Peppers meat seafood hot dogs bacon sausage 6
Blakes Chicken Parmesan Dinner frozen frozen meals 12
Bread, Healthy Whole Grain bakery bread 18
Brioche Bachelor Loaf bakery breakfast bakery 8
Brut RosÃ© alcohol specialty wines champagnes 8
Camilia, Single Liquid Doses other other 10
Chardonnay Paso Robles alcohol white wines 2
Chocolate Fudge Layer Cake frozen frozen dessert 8
Chocolate Sandwich Cookies snacks cookies cakes 152
Citrus Terere Yerba Mate beverages tea 4
Classic Culinary Low Sodium Vegetable Broth canned goods soup broth bouillon 4
Classics Earl Grey Tea beverages tea 20
Coconut Chocolate Chip Energy Bar snacks energy granola bars 6
Complete Spring Water Foaming Antibacterial Hand Wash personal care body lotions soap 14
Cookie Dough Performance Energy snacks energy granola bars 6
Corn Dogs, Mini, Honey Crunchy Flavor frozen frozen appetizers sides 4
Country Style Baked Beans canned goods canned meals beans 10
Cut Russet Potatoes Steam N' Mash frozen frozen produce 4
Total 13530
Filters
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Off
Keep all filters
On
product_name
is (All)
Report 1
Slicers & charts
Drill-Down
Details Page



The drill-through feature allows users to navigate from **Report 2** to a detailed view by selecting any value (e.g., a product) and using the **drill-through option**. Upon selection, the user is redirected to the **Details Page**, where comprehensive information related to the selected item is displayed.

For example, when selecting the product name **“Soda”**, the Details Page displays its corresponding **Department Name, Aisle Name, and Fact Order Count**, providing a clear and detailed view of its distribution.

product_name	department_name	aisle_name	Fact Order Count
Soda	beverages	soft drinks	7688
Total			**7688**